A Consolidated Overview of Artificial Neural Networks and Deep Neural Networks: Concepts, Architectures, and Training

Tobias Schlosser

Dresden University of Technology,

01307 Dresden, Germany,

{firstname.lastname}@tu-dresden.de

March 1, 2026

Abstract

Artificial neural networks (ANN) and deep neural networks (DNN) are core methodologies in modern machine learning, underpinning advances in computer vision, natural language processing, speech, and decision-making systems. This manuscript provides a consolidated overview of key concepts, architectural families, and training methods, with the goal of serving as a compact reference for researchers and practitioners. It summarizes foundational components such as neurons, activation and loss functions, regularization, initialization, and gradient-based learning, and it positions these elements within common modeling and evaluation workflows. The overview is structured into two main parts: (i) classical ANN concepts and architectures (including feed-forward, recurrent, and unsupervised formulations) and (ii) DNN-era architectures and paradigms (including CNNs, transformers, generative models, self-supervised learning, and large-scale training systems). Emphasis is placed on practical considerations that affect stability and generalization (e.g., normalization, learning-rate schedules, and gradient clipping), as well as on contemporary concerns such as efficiency, compression, robustness, and monitoring. The manuscript is intended as a living resource that will be extended as architectures and best practices evolve.

Table of Contents

1 Introduction and Motivation **5**

2 Artificial Neural Networks **6**

2.1 Neural building blocks and primitives . . . 7
2.1 Perceptron / artificial neuron . . . 7
2.2 Activation functions . . . 7
2.3 Loss functions . . . 7
2.4 Regularization (L2, dropout) . . . 7
2.5 Weight initialization . . . 7
2.2 Learning algorithms and optimization . . . 7
2.6 Backpropagation . . . 7
2.7 Stochastic gradient descent (SGD) . . . 7
2.8 Momentum / Nesterov . . . 7
2.9 Adam / adaptive methods . . . 7
2.10 Second-order methods . . . 7
2.11 Learning-rate schedules . . . 7
2.3 Generalization and model selection . . . 7
2.12 Early stopping . . . 7
2.13 Cross-validation . . . 7
2.14 Bias–variance trade-off . . . 7
2.15 Probability calibration . . . 7
2.4 Classic feed-forward networks . . . 8
2.16 Multilayer perceptron (MLP) . . . 8
2.17 Radial basis function network (RBF) . . . 8
2.18 Hopfield network . . . 8
2.5 Recurrent and sequence models (ANN era) . . . 8
2.19 Vanilla RNN . . . 8
2.20 LSTM . . . 8
2.21 GRU . . . 8
2.22 Seq2Seq (encoder–decoder) . . . 8
2.6 Associative, competitive, and unsupervised ANNs . . . 8
2.23 Self-organizing map (SOM) . . . 8
2.24 Competitive learning / LVQ . . . 8
2.25 Autoencoder (AE) . . . 8
2.26 Variational autoencoder (VAE) . . . 8
2.7 Stability, normalization, and training aids . . . 8
2.27 Batch normalization . . . 8
2.28 Layer normalization . . . 8
2.29 Gradient clipping . . . 9
2.30 Skip connections (concept) . . . 9
2.8 Evaluation and interpretability (general) . . . 9
2.31 Confusion-matrix metrics . . . 9
2.32 ROC/PR curves . . . 9
2.33 Sensitivity / saliency (basic) . . . 9

3 Deep Neural Networks **10**

3.1 Deep feed-forward and residual families . . . 11
3.1 Deep MLP stacks . . . 11
3.2 ResNet . . . 11
3.3 DenseNet . . . 11
3.4 EfficientNet . . . 11
3.2 Convolutional neural networks (CNN) . . . 11
3.5 LeNet-style CNN . . . 11
3.6 AlexNet . . . 11
3.7 VGG . . . 11
3.8 Inception / GoogLeNet . . . 11
3.9 MobileNet . . . 11
3.10 U-Net . . . 11
3.11 Fully convolutional networks (FCN) . . . 11
3.3 Sequence models and attention (deep era) . . . 11
3.12 Transformer . . . 11
3.13 BERT-style encoders . . . 11
3.14 GPT-style decoders . . . 11
3.15 Vision Transformer (ViT) . . . 11
3.16 Vision–language transformers . . . 11
3.4 Generative deep models . . . 12
3.17 Generative adversarial networks (GAN) . . . 12
3.18 StyleGAN family . . . 12
3.19 Diffusion models . . . 12
3.20 Autoregressive generative models . . . 12
3.21 Normalizing flows . . . 12
3.5 Self-supervised and representation learning . . . 12
3.22 Contrastive learning (SimCLR/MoCo) . . . 12
3.23 Masked modeling (MAE/MLM) . . . 12
3.24 Knowledge distillation . . . 12
3.6 Scaling, efficiency, and compression . . . 12
3.25 Mixed precision training . . . 12
3.26 Data parallelism . . . 12
3.27 Model/tensor parallelism . . . 12
3.28 Pipeline parallelism . . . 12
3.29 Pruning . . . 13
3.30 Quantization . . . 13
3.31 Low-rank adaptation / factorization . . . 13
3.7 Training paradigms and transfer . . . 13
3.32 Pretraining + fine-tuning . . . 13
3.33 Multi-task learning . . . 13
3.34 Domain adaptation . . . 13
3.35 Federated learning . . . 13
3.8 Robustness, safety, and monitoring . . . 13
3.36 Adversarial training . . . 13
3.37 Out-of-distribution detection . . . 13
3.38 Uncertainty estimation . . . 13
3.39 Drift monitoring . . . 13

3.9 Interpretability (deep models) 13
3.40 Grad-CAM 13
3.41 Integrated Gradients 13
3.42 SHAP (model-agnostic) 13

4 Conclusion and Outlook **14**

1 Introduction and Motivation

Artificial neural networks have evolved from simple perceptrons and multi-layer feed-forward models into deep architectures with billions of parameters and broad capabilities. This evolution has been driven by a combination of algorithmic innovations (e.g., effective backpropagation-based training, improved optimizers, normalization, and residual connections), increased computation capabilities, and large datasets. Today, ANNs are not only used as predictive models but also as general-purpose representation learners and generative systems, enabling transfer across tasks and modalities. As a result, the field has accumulated a large and sometimes fragmented set of terms, architectural patterns, and training heuristics that can be difficult to navigate.

A key challenge in practical work is that neural-network performance is shaped by many interacting choices beyond the architecture itself: data preprocessing, objective functions, regularization, initialization, optimization settings, and evaluation protocols can each dominate outcomes. Furthermore, modern deep learning introduces system-level considerations (e.g., distributed training, mixed precision, and memory constraints), while deployment adds further requirements (e.g., latency, robustness, monitoring, and interpretability). This manuscript consolidates essential building blocks and widely used architectural families into structured tables and accompanying text to support consistent terminology and more transparent experiment design.

Throughout the manuscript, ANN refers broadly to models trained from data via parameter optimization, while DNN denotes architectures with sufficient depth and capacity that benefit from modern training techniques and large-scale computation. The presented overview prioritizes concepts and families that are frequently encountered in contemporary literature and practice, and it aims to provide a baseline for deeper study rather than a historical or exhaustive treatment.

2 Artificial Neural Networks

Artificial neural networks can be understood as parameterized function approximators composed of simple computational units (neurons) arranged in layers or recurrent structures. At their core, ANNs combine linear transformations with nonlinear activation functions, and they are trained by minimizing a loss function using gradient-based optimization. Even in relatively shallow settings, these design choices determine representational capacity, optimization behavior, and generalization. Classical ANN practice therefore revolves around a small set of recurring elements: (i) choosing an objective aligned with the task (e.g., cross-entropy for classification or mean squared error for regression), (ii) stabilizing training via appropriate initialization and learning-rate control, and (iii) mitigating overfitting via regularization and validation-driven model selection.

Many architectural patterns commonly associated with modern deep learning have conceptual roots in the ANN era. Recurrent networks introduced parameter sharing across time and enabled sequential modeling, but also highlighted optimization pathologies such as vanishing and exploding gradients. Autoencoders and related unsupervised models established representation learning as a central objective, which later became crucial for self-supervised and foundation-model training. Similarly, normalization, gradient clipping, and skip connections can be viewed as training aids that address signal propagation and conditioning, allowing networks to scale in depth and to train more reliably. From an evaluation perspective, ANN workflows rely on well-understood classification and regression metrics, complemented by calibration and interpretability tools when probabilistic outputs or model explanations are needed.

Subsequently, the following abbreviations are used:

- AE = autoencoder
- ANN = artificial neural network
- DNN = deep neural network
- GRU = gated recurrent unit
- L2 = ℓ_2 (weight decay) regularization
- LSTM = long short-term memory
- LVQ = learning vector quantization
- MLP = multilayer perceptron
- MSE = mean squared error
- PR = precision–recall
- RBF = radial basis function
- ReLU = rectified linear unit
- ROC = receiver operating characteristic
- RNN = recurrent neural network
- Seq2Seq = sequence-to-sequence
- SGD = stochastic gradient descent
- SOM = self-organizing map
- VAE = variational autoencoder

Table 1: Artificial neural network concepts, components, and classical architectures.

Entry	Area	Modality	Scale	Typical tasks / notes
2.1 Neural building blocks and primitives				
Perceptron / artificial neuron [1]	Foundations	N/A	Core unit	Weighted sum + nonlinearity; binary/linear classification baseline
Activation functions [2]	Foundations	N/A	Core set	Sigmoid, tanh, ReLU family; affects gradient flow and expressivity
Loss functions [3]	Training	N/A	Core set	MSE, cross-entropy, hinge; objective shaping and calibration
Regularization (L2, dropout) [4]	Training	N/A	Core set	Controls overfitting; dropout as stochastic ensembling
Weight initialization [5]	Training	N/A	Core set	Xavier/Glorot, He; stabilizes forward/backward signal scales
2.2 Learning algorithms and optimization				
Backpropagation [6]	Training	N/A	Algorithm	Chain rule gradient computation; basis for most ANN/DNN training
Stochastic gradient descent (SGD) [7]	Optimization	N/A	Algorithm	Baseline optimizer; often paired with momentum
Momentum / Nesterov [8]	Optimization	N/A	Algorithm	Accelerates convergence; dampens oscillations
Adam / adaptive methods [9]	Optimization	N/A	Algorithm	Adaptive learning rates; widely used default in practice
Second-order methods [10]	Optimization	N/A	Algorithm	Newton/quasi-Newton; limited by compute/memory at scale
Learning-rate schedules [11]	Optimization	N/A	Technique	Step/cosine/warmup; crucial for stability and final performance
2.3 Generalization and model selection				
Early stopping [12]	Training	N/A	Technique	Stops training based on validation; strong regularizer
Cross-validation [13]	Evaluation	N/A	Protocol	Model selection for small/medium datasets; robust estimates
Bias–variance trade-off [14]	Theory	N/A	Concept	Explains under/overfitting; guides capacity control

continued on next page

Entry	Area	Modality	Scale	Typical tasks / notes
Probability calibration [15]	Evaluation	N/A	Technique	Reliability diagrams, temperature scaling; important for decisions
2.4 Classic feed-forward networks				
Multilayer perceptron (MLP) [16]	Architecture	Tabular/Text	Small–Large	Dense layers; baseline for tabular and learned embeddings
Radial basis function network (RBF) [17]	Architecture	Tabular	Small	Kernel-like hidden units; classic function approximation
Hopfield network [18]	Architecture	N/A	Small	Associative memory; energy-based formulation
2.5 Recurrent and sequence models (ANN era)				
Vanilla RNN [19]	Architecture	Sequence	Small–Medium	Sequential modeling; suffers from vanishing gradients
LSTM [20]	Architecture	Sequence	Medium–Large	Gated recurrence; long-range dependencies
GRU [21]	Architecture	Sequence	Medium–Large	Simplified gates vs LSTM; strong baseline
Seq2Seq (encoder–decoder) [22]	Architecture	Sequence	Medium–Large	Translation, summarization; often with attention
2.6 Associative, competitive, and unsupervised ANNs				
Self-organizing map (SOM) [23]	Unsupervised	Tabular	Small–Medium	Topology-preserving mapping; clustering/visualization
Competitive learning / LVQ Unsupervised	[24]	Tabular	Small	Prototype-based classification; interpretability via prototypes
Autoencoder (AE) [25]	Unsupervised	General	Small–Large	Representation learning; denoising and dimensionality reduction
Variational autoencoder (VAE) [26]	Generative	General	Medium–Large	Probabilistic latent variables; generation and disentanglement
2.7 Stability, normalization, and training aids				
Batch normalization [27]	Training	N/A	Technique	Stabilizes training; reduces sensitivity to initialization

continued on next page

Entry	Area	Modality	Scale	Typical tasks / notes
Layer normalization [28]	Training	N/A	Technique	Common in sequence models; normalization per sample
Gradient clipping [29]	Training	N/A	Technique	Prevents exploding gradients in RNNs and deep nets
Skip connections (concept) [30]	Architecture	N/A	Technique	Enables deeper networks; foundation for residual nets
2.8 Evaluation and interpretability (general)				
Confusion-matrix metrics [31]	Evaluation	N/A	Core set	Accuracy, precision/recall, F1; classification
ROC/PR curves [32]	Evaluation	N/A	Technique	Threshold analysis; PR preferred for imbalance
Sensitivity / saliency (basic) [33]	Interpretability	General	Technique	Input gradients; early neural explanation tools

3 Deep Neural Networks

Deep neural networks extend classical ANNs by scaling depth, width, and training data, and by introducing architectural and algorithmic mechanisms that improve optimization and transfer. In vision, convolutional neural networks (CNN) exploit locality and translational structure, yielding strong inductive biases for image-like data. Residual and densely connected families address optimization challenges by improving gradient flow and feature reuse, enabling much deeper networks. In sequence modeling, attention-based transformers have become the dominant architecture, replacing many recurrent approaches due to parallelizable training and strong scaling behavior. Across modalities, large-scale pretraining followed by task-specific fine-tuning (or prompting) has emerged as a default paradigm, shifting the focus from training from scratch to adapting general representations.

DNN practice is characterized by the tight coupling of modeling, training, and systems considerations. Stabilizing training at scale often requires normalization, carefully designed learning-rate schedules (including warmup), and techniques such as mixed precision to improve throughput. As model sizes grow, distributed training becomes essential, with data, model/tensor, and pipeline parallelism addressing different bottlenecks. At the same time, deployment constraints motivate compression methods such as pruning, quantization, and parameter-efficient adaptation (e.g., low-rank updates), which trade off accuracy, latency, memory, and energy. Robustness and safety concerns further motivate adversarial training, out-of-distribution detection, uncertainty estimation, and drift monitoring in production settings. Finally, interpretability remains an important companion to DNN adoption, especially in high-stakes domains, where attribution methods and monitoring help to detect spurious behavior and to support human oversight.

Subsequently, the following abbreviations are used:

- BF16 = brain floating point (bfloat16)
- CNN = convolutional neural network
- FP16 = 16-bit floating point
- GAN = generative adversarial network
- Grad-CAM = gradient-weighted class activation mapping
- INT8/INT4 = 8-/4-bit integer quantization
- LoRA = low-rank adaptation
- MAE = masked autoencoder
- MLM = masked language modeling
- MLOps = machine-learning operations
- MLP = multilayer perceptron
- OOD = out-of-distribution
- ResNet = residual network
- SHAP = SHapley Additive exPlanations
- V+L = vision–language
- ViT = vision transformer

Table 2: Deep neural network architectures, training paradigms, and system considerations.

Entry	Area	Modality	Scale	Typical tasks / notes
3.1 Deep feed-forward and residual families				
Deep MLP stacks [34]	Architecture	Tabular/Text	Medium–Large	Depth increases expressivity; sensitive to init/normalization
ResNet [30]	Architecture	Vision	Large	Residual blocks; reliable backbone for many tasks
DenseNet [35]	Architecture	Vision	Medium–Large	Dense connectivity; feature reuse; memory considerations
EfficientNet [36]	Architecture	Vision	Large	Compound scaling of depth/width/resolution; efficient accuracy
3.2 Convolutional neural networks (CNN)				
LeNet-style CNN [37]	Architecture	Vision	Small	Early CNN; pedagogy and simple baselines
AlexNet [38]	Architecture	Vision	Medium	Breakthrough large-scale CNN; historical reference
VGG [39]	Architecture	Vision	Large	Deep uniform conv stacks; heavy compute/memory
Inception / GoogLeNet [40]	Architecture	Vision	Large	Multi-branch modules; efficient receptive fields
MobileNet [41]	Architecture	Vision	Small–Medium	Depthwise separable conv; mobile/edge deployment
U-Net [42]	Architecture	Vision/MedImg	Medium	Encoder–decoder with skip connections; segmentation staple
Fully convolutional networks (FCN) [43]	Architecture	Vision	Medium	Dense prediction; semantic segmentation groundwork
3.3 Sequence models and attention (deep era)				
Transformer [44]	Architecture	Text/Vision	Large	Self-attention; foundation model backbone
BERT-style encoders [45]	Architecture	Text	Large	Masked-language pretraining; NLU and embeddings
GPT-style decoders [46]	Architecture	Text	Large	Autoregressive generation; prompting and instruction tuning
Vision Transformer (ViT) [47]	Architecture	Vision	Large	Patch embeddings + attention; benefits from large-scale pretraining

continued on next page

Entry	Area	Modality	Scale	Typical tasks / notes
Vision–language transformers [44]	Architecture	V+L	Large	Image-text alignment; retrieval and captioning
3.4 Generative deep models				
Generative adversarial networks (GAN) [48]	Generative	General	Medium–Large	Adversarial training; sharp images; training instability
StyleGAN family [49]	Generative	Vision	Large	High-fidelity face/image synthesis; controllable styles
Diffusion models [50]	Generative	Vision/Audio	Large	Score-based/denoising diffusion; strong fidelity and diversity
Autoregressive generative models [51]	Generative	Text/Vision	Large	Token/patch generation; likelihood-based training
Normalizing flows [52]	Generative	General	Medium	Exact likelihood; invertible transformations; memory/compute trade-offs
3.5 Self-supervised and representation learning				
Contrastive learning (SimCLR/MoCo) [53, 54]	Self-supervised	Vision	Large	Augmentation-driven invariances; strong pretraining
Masked modeling (MAE/MLM) [45, 55]	Self-supervised	Text/Vision	Large	Reconstruction of masked tokens/patches
Knowledge distillation [56]	Training	General	Medium–Large	Teacher–student compression; accuracy vs latency trade-offs
3.6 Scaling, efficiency, and compression				
Mixed precision training [57]	Systems	General	Large	FP16/BF16; faster training; care with stability
Data parallelism [58]	Systems	General	Large	Scale batch size across devices; communication overhead
Model/tensor parallelism [59]	Systems	General	Large	Splits model across devices; necessary for huge models

continued on next page

Entry	Area	Modality	Scale	Typical tasks / notes
Pipeline parallelism [60]	Systems	General	Large	Stage-wise splitting; bubble/latency considerations
Pruning [61]	Compression	General	Medium–Large	Unstructured/structured pruning; latency gains vary
Quantization [62]	Compression	General	Small–Large	INT8/INT4; deployment efficiency; calibration required
Low-rank adaptation / factorization [63]	Compression	General	Medium–Large	LoRA-style adapters; parameter-efficient finetuning
3.7 Training paradigms and transfer				
Pretraining + fine-tuning [64]	Training	General	Large	Dominant paradigm for modern DNNs
Multi-task learning [65]	Training	General	Medium–Large	Shared representations across tasks; weighting challenges
Domain adaptation [66]	Training	General	Medium	Shift handling; covariate/label shift strategies
Federated learning [67]	Training	Distributed	Medium–Large	Privacy-aware training across sites; non-IID issues
3.8 Robustness, safety, and monitoring				
Adversarial training [68]	Robustness	General	Medium–Large	Improves robustness; computationally expensive
Out-of-distribution detection [69]	Robustness	General	Medium	Shift detection; uncertainty and density proxies
Uncertainty estimation [70]	Robustness	General	Medium	Calibration, ensembles, Bayesian approximations
Drift monitoring [71]	MLOps	General	Production	Detects data/label drift; triggers retraining/alerts
3.9 Interpretability (deep models)				
Grad-CAM [72]	Interpretability	Vision	Medium	Class-discriminative localization for CNNs
Integrated Gradients [73]	Interpretability	General	Medium	Axiomatic attribution; works with differentiable models
SHAP (model-agnostic) [74]	Interpretability	General	Medium	Feature attributions; often expensive for deep models

4 Conclusion and Outlook

This manuscript consolidates core concepts, architectural families, and training methods for artificial and deep neural networks into a structured overview. The ANN section emphasizes foundational primitives (activations, losses, regularization, and initialization) and classical architectures that remain relevant as baselines and conceptual building blocks. The DNN section highlights modern architectural families (CNNs, residual networks, and transformers), generative modeling approaches, self-supervised pretraining, and the system-level techniques required to train and deploy large models. Across both, recurring practical themes emerged: stable optimization depends on well-chosen learning rates and normalization; generalization depends on capacity control, validation protocols, and calibration; and real-world deployment requires efficiency, robustness, and monitoring.

Looking forward, several developments are likely to shape neural-network practice. First, evaluation will increasingly incorporate multiple objectives beyond accuracy, including robustness under distribution shift, calibration, fairness, efficiency, and environmental cost. Second, pretraining-driven workflows will continue to blur modality boundaries, placing greater emphasis on multimodal architectures, data governance, and contamination-aware benchmarking. Third, efficient adaptation and compression will become standard as models move from research to constrained deployment environments, motivating continued work on quantization, pruning, and parameter-efficient fine-tuning. Finally, trustworthy deployment will require tighter integration of interpretability, uncertainty estimation, and drift monitoring into MLOps pipelines.

References

[1] F. Rosenblatt, "The perceptron: a probabilistic model for information storage and organization in the brain." *Psychological review*, vol. 65, no. 6, p. 386, 1958.
(1 citation on 1 page: 7)

[2] A. Apicella, F. Donnarumma, F. Isgrò, and R. Prevete, "A survey on modern trainable activation functions," *Neural Networks*, vol. 138, pp. 14–32, 2021.
(1 citation on 1 page: 7)

[3] Q. Wang, Y. Ma, K. Zhao, and Y. Tian, "A comprehensive survey of loss functions in machine learning," *Annals of Data Science*, vol. 9, no. 2, pp. 187–212, 2022.
(1 citation on 1 page: 7)

[4] R. Moradi, R. Berangi, and B. Minaei, "A survey of regularization strategies for deep models," *Artificial Intelligence Review*, vol. 53, no. 6, pp. 3947–3986, 2020.
(1 citation on 1 page: 7)

[5] M. V. Narkhede, P. P. Bartakke, and M. S. Sutaone, "A review on weight initialization strategies for neural networks," *Artificial intelligence review*, vol. 55, no. 1, pp. 291–322, 2022.
(1 citation on 1 page: 7)

[6] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *nature*, vol. 323, no. 6088, pp. 533–536, 1986.
(1 citation on 1 page: 7)

[7] H. Robbins and S. Monro, "A stochastic approximation method," *The annals of mathematical statistics*, pp. 400–407, 1951.
(1 citation on 1 page: 7)

[8] Y. Nesterov, "A method for solving the convex programming problem with convergence rate o (1/k2)," in *Dokl akad nauk Sssr*, vol. 269, 1983, p. 543.
(1 citation on 1 page: 7)

[9] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
(1 citation on 1 page: 7)

[10] D. C. Liu and J. Nocedal, "On the limited memory bfgs method for large scale optimization," *Mathematical programming*, vol. 45, no. 1, pp. 503–528, 1989.
(1 citation on 1 page: 7)

[11] J. Konar, P. Khandelwal, and R. Tripathi, "Comparison of various learning rate scheduling techniques on convolutional neural network," in *2020 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECS)*. IEEE, 2020, pp. 1–5.
(1 citation on 1 page: 7)

[12] L. Prechelt, "Early stopping-but when?" in *Neural Networks: Tricks of the trade*. Springer, 2002, pp. 55–69.
(1 citation on 1 page: 7)

[13] M. Stone, "Cross-validatory choice and assessment of statistical predictions," *Journal of the royal statistical society: Series B (Methodological)*, vol. 36, no. 2, pp. 111–133, 1974.
(1 citation on 1 page: 7)

[14] S. Geman, E. Bienenstock, and R. Doursat, "Neural networks and the bias/variance dilemma," *Neural computation*, vol. 4, no. 1, pp. 1–58, 1992.
(1 citation on 1 page: 7)

[15] W. B. Glenn *et al.*, "Verification of forecasts expressed in terms of probability," *Monthly weather review*, vol. 78, no. 1, pp. 1–3, 1950.
(1 citation on 1 page: 8)

[16] F. Rosenblatt *et al.*, *Principles of neurodynamics: Perceptrons and the theory of brain mechanisms*. Spartan books Washington, DC, 1962, vol. 55.
(1 citation on 1 page: 8)

[17] D. S. Broomhead and D. Lowe, "Radial basis functions, multi-variable functional interpolation and adaptive networks," Tech. Rep., 1988.
(1 citation on 1 page: 8)

[18] J. J. Hopfield, "Neural networks and physical systems with emergent collective computational abilities." *Proceedings of the national academy of sciences*, vol. 79, no. 8, pp. 2554–2558, 1982.
(1 citation on 1 page: 8)

[19] W. S. McCulloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity," *The bulletin of mathematical biophysics*, vol. 5, no. 4, pp. 115–133, 1943.
(1 citation on 1 page: 8)

[20] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
(1 citation on 1 page: 8)

[21] K. Cho, B. Van Merriënboer, Ç. Gulçehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using rnn encoder–decoder for statistical machine translation," in *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, 2014, pp. 1724–1734.
(1 citation on 1 page: 8)

[22] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," *Advances in neural information processing systems*, vol. 27, 2014.
(1 citation on 1 page: 8)

[23] T. Kohonen, "Self-organized formation of topologically correct feature maps," *Biological cybernetics*, vol. 43, no. 1, pp. 59–69, 1982.
(1 citation on 1 page: 8)

[24] ——, *Self-organization and associative memory*. Springer Science & Business Media, 2012.
(1 citation on 1 page: 8)

[25] D. E. Rumelhart, J. L. McClelland, P. R. Group *et al.*, *Parallel distributed processing, volume 1: Explorations in the microstructure of cognition: Foundations*. The MIT press, 1986.
(1 citation on 1 page: 8)

[26] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," *arXiv preprint arXiv:1312.6114*, 2013.
(1 citation on 1 page: 8)

[27] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *International conference on machine learning*. pmlr, 2015, pp. 448–456.
(1 citation on 1 page: 8)

[28] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.
(1 citation on 1 page: 9)

[29] R. Pascanu, T. Mikolov, and Y. Bengio, "On the difficulty of training recurrent neural networks," in *International conference on machine learning*. Pmlr, 2013, pp. 1310–1318.
(1 citation on 1 page: 9)

[30] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
(2 citations on 2 pages: 9 and 11)

[31] G. A. Miller and P. E. Nicely, "An analysis of perceptual confusions among some english consonants," *The Journal of the Acoustical Society of America*, vol. 27, no. 2, pp. 338–352, 1955.
(1 citation on 1 page: 9)

[32] W. Peterson, T. Birdsall, and W. Fox, "The theory of signal detectability," *Transactions of the IRE professional group on information theory*, vol. 4, no. 4, pp. 171–212, 1954.
(1 citation on 1 page: 9)

[33] C. Koch and S. Ullman, "Shifts in selective visual attention: towards the underlying neural circuitry." *Human neurobiology*, vol. 4, no. 4, pp. 219–227, 1985.
(1 citation on 1 page: 9)

[34] I. O. Tolstikhin, N. Houlsby, A. Kolesnikov, L. Beyer, X. Zhai, T. Unterthiner, J. Yung, A. Steiner, D. Keysers, J. Uszkoreit *et al.*, "Mlp-mixer: An all-mlp architecture for vision," *Advances in neural information processing systems*, vol. 34, pp. 24 261–24 272, 2021.
(1 citation on 1 page: 11)

[35] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4700–4708.
(1 citation on 1 page: 11)

[36] M. Tan and Q. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *International conference on machine learning*. PMLR, 2019, pp. 6105–6114.
(1 citation on 1 page: 11)

[37] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 2002.
(1 citation on 1 page: 11)

[38] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.
(1 citation on 1 page: 11)

[39] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.
(1 citation on 1 page: 11)

[40] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 1–9.
(1 citation on 1 page: 11)

[41] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017.
(1 citation on 1 page: 11)

[42] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.
(1 citation on 1 page: 11)

[43] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 3431–3440.
(1 citation on 1 page: 11)

[44] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.
(2 citations on 2 pages: 11 and 12)

[45] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, 2019, pp. 4171–4186.
(2 citations on 2 pages: 11 and 12)

[46] A. Radford, K. Narasimhan, T. Salimans, I. Sutskever *et al.*, "Improving language understanding by generative pre-training," 2018.
(1 citation on 1 page: 11)

[47] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.
(1 citation on 1 page: 11)

[48] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," *Advances in neural information processing systems*, vol. 27, 2014.
(1 citation on 1 page: 12)

[49] T. Karras, S. Laine, and T. Aila, "A style-based generator architecture for generative adversarial networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 4401–4410.
(1 citation on 1 page: 12)

[50] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in *International conference on machine learning*. pmlr, 2015, pp. 2256–2265.
(1 citation on 1 page: 12)

[51] A. Van Den Oord, N. Kalchbrenner, and K. Kavukcuoglu, "Pixel recurrent neural networks," in *International conference on machine learning*. PMLR, 2016, pp. 1747–1756.
(1 citation on 1 page: 12)

[52] D. Rezende and S. Mohamed, "Variational inference with normalizing flows," in *International conference on machine learning*. PMLR, 2015, pp. 1530–1538.
(1 citation on 1 page: 12)

[53] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *International conference on machine learning*. PmLR, 2020, pp. 1597–1607.
(1 citation on 1 page: 12)

[54] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 9729–9738.
(1 citation on 1 page: 12)

[55] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 16 000–16 009.
(1 citation on 1 page: 12)

[56] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.
(1 citation on 1 page: 12)

[57] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh *et al.*, "Mixed precision training," *arXiv preprint arXiv:1710.03740*, 2017.
(1 citation on 1 page: 12)

[58] J. Dean, G. Corrado, R. Monga, K. Chen, M. Devin, M. Mao, M. Ranzato, A. Senior, P. Tucker, K. Yang *et al.*, "Large scale distributed deep networks," *Advances in neural information processing systems*, vol. 25, 2012.
(1 citation on 1 page: 12)

[59] M. Shoeybi, M. Patwary, R. Puri, P. LeGresley, J. Casper, and B. Catanzaro, "Megatron-lm: Training multi-billion parameter language models using model parallelism," *arXiv preprint arXiv:1909.08053*, 2019.
(1 citation on 1 page: 12)

[60] Y. Huang, Y. Cheng, A. Bapna, O. Firat, D. Chen, M. Chen, H. Lee, J. Ngiam, Q. V. Le, Y. Wu *et al.*, "Gpipe: Efficient training of giant neural networks using pipeline parallelism," *Advances in neural information processing systems*, vol. 32, 2019.
(1 citation on 1 page: 13)

[61] Y. LeCun, J. Denker, and S. Solla, "Optimal brain damage," *Advances in neural information processing systems*, vol. 2, 1989.
(1 citation on 1 page: 13)

[62] M. Courbariaux, Y. Bengio, and J.-P. David, "Binaryconnect: Training deep neural networks with binary weights during propagations," *Advances in neural information processing systems*, vol. 28, 2015.
(1 citation on 1 page: 13)

[63] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen *et al.*, "Lora: Low-rank adaptation of large language models." *Iclr*, vol. 1, no. 2, p. 3, 2022.
(1 citation on 1 page: 13)

[64] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *science*, vol. 313, no. 5786, pp. 504–507, 2006.
(1 citation on 1 page: 13)

[65] R. Caruana, "Multitask learning," *Machine learning*, vol. 28, no. 1, pp. 41–75, 1997.
(1 citation on 1 page: 13)

[66] X. Glorot, A. Bordes, and Y. Bengio, "Domain adaptation for large-scale sentiment classification: A deep learning approach," in *Proceedings of the 28th international conference on machine learning (ICML-11)*, 2011, pp. 513–520.
(1 citation on 1 page: 13)

[67] J. Konečnỳ, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, and D. Bacon, "Federated learning: Strategies for improving communication efficiency," *arXiv preprint arXiv:1610.05492*, 2016.
(1 citation on 1 page: 13)

[68] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," *arXiv preprint arXiv:1312.6199*, 2013.
(1 citation on 1 page: 13)

[69] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," *arXiv preprint arXiv:1610.02136*, 2016.
(1 citation on 1 page: 13)

[70] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in *international conference on machine learning*. PMLR, 2016, pp. 1050–1059.
(1 citation on 1 page: 13)

[71] S. Greco, B. Vacchetti, D. Apiletti, and T. Cerquitelli, "Unsupervised concept drift detection from deep learning representations in real-time," *IEEE Transactions on Knowledge and Data Engineering*, 2025.
(1 citation on 1 page: 13)

[72] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 618–626.
(1 citation on 1 page: 13)

[73] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *International conference on machine learning*. PMLR, 2017, pp. 3319–3328.
(1 citation on 1 page: 13)

[74] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in neural information processing systems*, vol. 30, 2017.
(1 citation on 1 page: 13)